

A MARGINAL REGRESSION MODEL FOR LONGITUDINAL COMPOSITIONAL COUNTS WITH APPLICATION TO MICROBIOME DATA

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Infant gut microbial compositions are disrupted by early-life exposures and constantly change in the first few years of life. Association analyses are performed to study the effects of early-life exposures on the trajectories of microbiota maturation. Microbiome data from sequencing experiments contain compositional counts of various microbial taxa that exhibit varying levels of zero inflation and overdispersion. There is a lack of methods to model compositional counts with these characteristics. To address this challenge, we first propose a distribution named adaptively zero-inflated generalized Dirichlet multinomial (AIGDM) that uses GDM to model the relative abundance of the present taxa and the zero-inflation part to model taxa absence when needed. We introduce a likelihood-ratio test to determine the necessity of having the zero-inflation part for each taxon. We then develop an AIGDM-based marginal regression model for longitudinal microbiome compositional counts. The model combines the ability of AIGDM to flexibly model microbial compositions and the ability of the generalized estimating equation method (GEE) to handle correlations between the repeated measures. Under the model, we propose association tests for mean, dispersion, and absence-presence proportion parameters to characterize what aspect of the microbial composition distribution is disrupted by the exposure in a longitudinal study. We also propose an omnibus test by combining these tests to achieve overall power and robustness. Through extensive simulation studies, we demonstrate the advantages of AIGDM tests over existing association tests for microbial compositions. We showcase the power of the AIGDM tests by analyzing the dataset of an infant gut microbiome longitudinal study and identifying differential microbial compositions between C-section and vaginal-birth infants at various taxonomic levels.

1. Introduction. The human microbiome, a collection of microorganisms that reside in and on the human body, has been established to play a significant role in many biological functions of human life (Gilbert et al., 2018; The Integrative HMP (iHMP) Research Network Consortium, 2019). The longitudinal design has been adopted in an increasing number of microbiome studies to investigate the temporal variability of microbial compositions and their potential biomedical relevance. In longitudinal microbiome studies, scientists often compare microbial composition trajectories between different groups. Our work is motivated by the prominent Early Childhood and the Microbiome (ECAM) study (Bokulich et al., 2016) that profiled microbial development during the first two years of life in a cohort of 43 US infants. The goal of this study is to evaluate how early life exposures, such as delivery mode, influence the childhood microbial community development.

Microbiome data are usually generated from high-throughput sequencing experiments. The sequencing reads are assigned to individual operational taxonomic units (OTUs) or amplicon sequence variants (ASVs), which are further classified into taxa at different taxonomic levels (e.g., genus, family, order, class, phylum, and kingdom) (Bolyen et al., 2019). Because the sequencing depth (the total number of reads obtained for a sample) is not correlated with

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the actual microbial load (the total amount of microbial DNA) in the sample, the data are compositional counts representing the relative abundances of various taxa and their absolute abundance cannot be recovered from the sequencing experiment. The relative abundance of all taxa would shift even when only one taxon changes its underlying absolute abundance. Moreover, microbiome data are highly sparse, and many taxa exhibit zero inflation patterns in abundance distribution (Kaul et al., 2017). Zero counts may arise from the absence of the taxa in the samples (referred to as true zeros) or from the undersampling of the taxa (referred to as sampling zeros). In addition, the microbial composition exhibits high degree of overdispersion due to heterogeneity among subjects and the complex correlation among taxa. These characteristics of microbiome compositional counts make them challenging to model using standard distribution for compositional counts (e.g., multinomial). Association analysis methods ignoring the key data characteristics can lose power and yield a high false discovery rate (Weiss et al., 2015; Gloor et al., 2017; Hawinkel et al., 2017).

QCATC (Tang and Chen, 2021) is a distribution-free composition association test for correlated microbiome samples. The method makes no distributional assumption on the compositional counts and is robust against distribution misspecification. The two-part version of QCATC models positive and zero counts separately and combines association evidence from both parts. The two-part test substantially gains power when testing compositions with excessive zero counts.

A parametric test is well-powered when it adopts a distribution that appropriately addresses the data characteristics. An oversimple distribution misses the key data characteristics, and an overcomplex distribution wastes parameters on granular details, which adversely affects the statistical efficiency. Some differential composition tests assume the compositional counts follow the Dirichlet multinomial (DM) distribution (La Rosa et al., 2012). In the DM, the counts are sampled from the multinomial distribution with underlying proportion parameters following a Dirichlet distribution. However, DM is often too simple for microbiome compositions (Holmes et al., 2012; Silverman et al., 2020). DM has only one overdispersion parameter and fails to flexibly account for the diverse dispersion and zero-inflation levels for all taxa in the composition. Moreover, DM always imposes negative correlations among taxa, whereas the actual data display positive and negative correlations (Li, 2015). The generalized DM (GDM) distribution (Connor and Mosimann, 1969; Wong, 1998) is an extension of DM by allowing the taxon-specific dispersion model and a more flexible correlation structure among taxa. It has been demonstrated that GDM fits the real microbiome data better than DM in most scenarios (Tang and Chen, 2019). Another distribution for microbiome compositional count is logistic normal multinomial (LNM) (Xia et al., 2013). This distribution can flexibly model overdispersion and correlation among taxa. However, the LNM-based analysis is more computationally intensive because the logistic normal is not a conjugate prior for multinomial.

The distributions mentioned above do not model taxa absence (i.e., the underlying proportion parameters cannot be exactly zero), and all observed zero counts are considered as the sampling zeros. Zero-inflated GDM (ZIGDM) was introduced to overcome this limitation (Tang and Chen, 2019). The zero inflation part of the ZIGDM is for modeling true zeros and accounting for the excessive observed zero counts that cannot be explained by the sampling zeros in the GDM model. However, the ZIGDM assumes zero inflation for all taxa, which could be superfluous for some taxa.

To determine the necessity of zero-inflation modeling, we propose a likelihood ratio test for each taxon under the GDM null model and investigate if adding the zero-inflation significantly improves the goodness-of-fit. The GDM-based distribution with zero inflation adaptively determined this way is referred to as AIGDM. Based on AIGDM, we introduce a marginal regression model to analyze longitudinal microbiome compositional count

data. To efficiently estimate parameters in the AIGDM marginal model, we develop an expectation-solution (ES) algorithm (Rosen et al., 2000) that combines elements of the expectation-maximization (EM) algorithm and the generalized estimating equations (GEE) method (Zeger et al., 1988). By virtue of GEE, statistical inference accounts for the correlation between repeated measures within a subject, even if the working correlation structure is incorrectly specified. Under the marginal regression model, we perform three score tests to detect differential mean, dispersion, and absence/presence proportion of taxa in the composition. We then construct an omnibus test by combining these test p -values to achieve robust and overall high power.

The remainder of this article is structured as follows. Section 2 introduces the motivating microbiome dataset and the scientific question of interest. Section 3 presents details of the AIGDM marginal model and the association tests. Section 4 presents simulation studies. Section 5 presents the data analysis results for the motivating dataset. Section 6 concludes the article with a discussion.

2. ECAM: Early Childhood and Microbiome Study. During the initial years of life, the gut microbial community undergoes significant changes, evolving and maturing until it reaches an adult-like state. Early-life exposure, such as C-section delivery, can disrupt this delicate balance and potentially lead to long-term health implications. To profile microbial development during infancy and depict the disturbances associated with early-life exposures, the ECAM study (Bokulich et al., 2016) longitudinally measured the gut microbiome of 43 US infants at regular intervals from birth to 2 years old. Among these infants, 19 are C-section birth and 24 are vaginal birth. The sampled time points for each infant are shown in Figure 1a, producing a total of 840 samples.

Microbiome samples were sequenced by 16S rRNA sequencing platform. Raw data were downloaded from Qiita (<https://qiita.ucsd.edu/>) (Study ID: 10249). The compositional count of OTUs and their identities at different taxonomic ranks were generated by QIIME2 pipeline (Bolyen et al., 2019). The original analysis (Bokulich et al., 2016) of the ECAM data revealed that gut microbial diversity is different between vaginal birth and C-section birth infants. Figure 1b shows that infants delivered vaginally have a steadily increased and higher phylogenetic diversity in their gut microbiome than infants born via C-section. A more in-depth analysis to discover groups of taxa that contribute to this community-level difference is of great scientific interest. The original analysis employed a univariate differential abundance test LEfSe (Segata et al., 2011) to analyze the relative abundance of each taxon. Such analysis ignores the compositional nature of microbiome data and is prone to an excessive number of false discoveries (Gloor et al., 2017; Hawinkel et al., 2017).

We approach this question by analyzing compositions defined on the taxonomy tree. Incorporating information on the evolutionary relationship among taxa into the analysis of microbiome data can potentially improve the statistical power (Tang et al., 2017; Washburne et al., 2018; Hong et al., 2023) because the taxa that share a certain degree of evolutionary affinity tend to have similar biological functions. The left panel in Figure 1c displays a taxonomy tree for taxa in the ECAM data. The tree has six levels from kingdom to genus (the outer circle represents a lower taxonomic level). At each internal node, the child taxa at the lower rank form a composition. The compositions defined this way through all internal nodes contain groups of evolutionarily close taxa at various taxonomic resolutions. Our analysis aims to detect significant differences in the trajectories of these compositions over the first two years of life between infants of different delivery modes (an example of composition trajectories is shown in the right panel of Figure 1c). Additionally, our analysis provides insight into which aspect of the relative abundance distribution (mean, dispersion, proportion of absence/presence) differs.

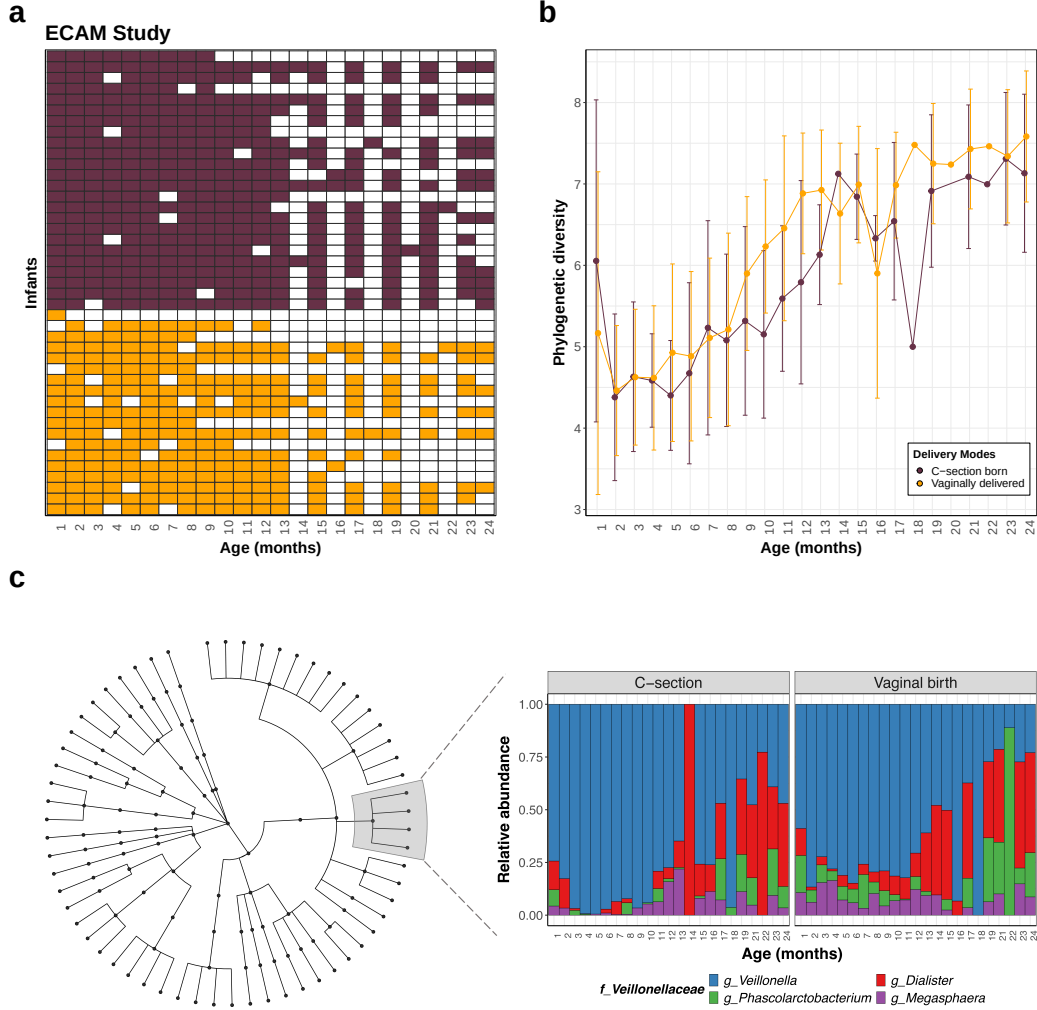


Fig 1: Data of the ECAM study. (a) Each infant's sampling time points over the two years with different colors representing C-section birth (brown) and vaginal birth (orange). More than one sample can be generated each month. (b) The trajectory of phylogenetic diversity over the two years for the C-section and vaginally delivered infants. Points represent mean values, and error bars encapsulate the mean $\pm$ standard deviation range. (c) Illustration of a taxonomic tree from kingdom (root node at the center) to genus (terminal nodes at the outmost circle) and the trajectory of the relative abundance of a genus composition under family *Veillonellaceae* over the two years for the two birth groups.

3. Methodology. We formally present the AIGDM marginal model and association tests. In Section 3.1, we briefly review the ZIGDM distribution. In Section 3.2, we introduce a likelihood ratio test to determine whether modeling zero inflation for each taxon is necessary. In Section 3.3, we develop the AIGDM marginal model for longitudinal compositional count data and the ES algorithm for parameter estimation. In Section 3.4, we develop score tests for composition association analysis under the AIGDM marginal model.

3.1. ZIGDM Distribution. Consider a microbial composition consisting of $K + 1$ taxa. Let Y_j denote the compositional count of taxon j . It is natural to assume that $\mathbf{Y} = (Y_1, \dots, Y_{K+1})$ follows a multinomial distribution with size $\mathbb{N} = \sum_{j=1}^{K+1} Y_j$ and proportion parameters $\mathbf{P} = (P_1, \dots, P_{K+1})$ with the unit-sum constraint $\sum_{j=1}^{K+1} P_j = 1$. To address the overdispersion of microbial compositions, we treat $\mathbf{P}$ as a random vector. By assuming $\mathbf{P}$ follows zero-inflated generalized Dirichlet (ZIGD), the marginal distribution of $\mathbf{Y}$ follows ZIGDM (Tang and Chen, 2019). The ZIGD is a flexible distribution to accommodate overdispersion, zero-inflation, and the complex correlation structure among taxa. The ZIGD sample can be constructed from a stick-breaking process (Ishwaran and James, 2001) with modification that permits zero components in the proportions for modeling taxa absence. Specifically, the stick-breaking process starts with an initial stick of length 1. At each step, a fraction of the remaining stick is broken off randomly, with the broken length following a zero-inflated beta (ZIB) distribution. The process continues recursively with the remaining part of the stick until $K + 1$ components are reached. Taken together, the mathematical formula for generating $\mathbf{Y}$ can be expressed in the following hierarchical model

$$(3.1) \quad \begin{aligned} \Delta_j &\sim \text{Bernoulli}(\pi_j), \quad j = 1, \dots, K \\ Z_j &= 0 \text{ if } \Delta_j = 1; \quad (Z_j \mid \Delta_j = 0) \sim \text{Beta}(a_j, b_j) \\ P_1 &= Z_1, \quad P_j = Z_j \prod_{k=1}^{j-1} (1 - Z_k), \quad j = 2, \dots, K, \quad P_{K+1} = 1 - \sum_{k=1}^K P_k \\ (\mathbf{Y} \mid \mathbf{P}, \mathbb{N}) &\sim \text{Multinomial}(\mathbf{P}, \mathbb{N}), \end{aligned}$$

where Δ_j is a latent variable with the value of 0 or 1 indicating the presence or absence of taxon j , π_j controls the absence probability, and $Z_1, \dots, Z_K$ are K independent ZIB variables, each characterized by two shape parameters a_j and b_j . The mean of the beta distribution is $\mu_j = a_j / (a_j + b_j)$ and variance is $\phi_j \mu_j (1 - \mu_j)$, where $\phi_j = 1 / (a_j + b_j + 1)$ is a dispersion parameter characterizing the spread of the distribution. By treating Δ_j 's and Z_j 's as latent variables, the parameters in the ZIGDM model can be efficiently estimated by an EM algorithm (Tang and Chen, 2019). The ZIGDM reduces to GDM if we fix $\Delta_j = 0$ for all taxa.

3.2. Zero-Inflation Determination. The ZIGDM assumes zero inflation for all taxa, whereas GDM does not assume zero inflation for any taxa. These two extreme scenarios may not reflect the reality of real data where some taxa exhibit zero inflation, but others do not. Whether or not modeling zero inflation affects the power of association tests (Tang and Chen, 2019). Therefore, we propose first performing a test to diagnose the existence of zero inflation for every taxon and then only model zero inflation for the taxa whose zero counts cannot be fully explained by the background GDM distribution.

For a ZIGDM composition $\mathbf{Y}$ formulated in (3.1), conditional on $\mathbb{N}_j = \sum_{k=j}^{K+1} Y_k$, each component Y_j follows a zero-inflated beta-binomial (ZIBB) distribution with parameters π_j , a_j , and b_j . Therefore, we can test for zero inflation for individual taxa separately by comparing the goodness-of-fit of a beta-binomial (BB) against a ZIBB.

Suppose we have n samples obtained from longitudinal measurements of individual subjects. Let Y_{ij} denote the count of taxon j in sample i . For taxon j , we test the hypothesis $H_0 : (Y_{ij} \mid \mathbb{N}_{ij}) \sim \text{BB}$ against $H_1 : (Y_{ij} \mid \mathbb{N}_{ij}) \sim \text{ZIBB}$ to determine whether zero-inflation model is needed. We propose to use the pseudo likelihood ratio test (LRT) with the test statistic given by

$$\Lambda = \log \frac{\prod_{i=1}^n f_{\text{BB}}(Y_{ij}; \mathbb{N}_{ij}, \tilde{a}_j, \tilde{b}_j)}{\prod_{i=1}^n f_{\text{ZIBB}}(Y_{ij}; \mathbb{N}_{ij}, \hat{\pi}_j, \hat{a}_j, \hat{b}_j)},$$

where $f_{\text{BB}}(Y_{ij}; \mathbb{N}_{ij}, \tilde{a}_j, \tilde{b}_j)$ is the density function of BB with the pseudo maximum likelihood estimators (MLE) $\tilde{a}_j$ and $\tilde{b}_j$ obtained by fitting BB to data $(Y_{ij} \mid \mathbb{N}_{ij})$, and

$f_{\text{ZIBB}}(Y_{ij}; \mathbb{N}_{ij}, \hat{\pi}_j, \hat{a}_j, \hat{b}_j)$ is the density function of ZIBB with the pseudo MLE $\hat{\pi}_j$, $\hat{a}_j$ and $\hat{b}_j$ obtained by fitting ZIBB to the same data. Here, the term “pseudo” means that we treat individual samples as if they were independent. In light of previous literature on statistical inference for the number of components in a mixture model (Lo et al., 2001; Aldirawi et al., 2019), we propose a bootstrapped Monte Carlo in Algorithm 1 to estimate the p -value of the LRT.

Algorithm 1 Estimating LRT p -value

Given the data $\mathbf{D} = \{(Y_{ij} | \mathbb{N}_{ij})\}_{i=1}^n$ $\ell = 1, \dots, B$
 Resample $\mathbf{D}$ at the subject-level with replacement to obtain a bootstrapped data $\mathbf{D}^{(\ell)} = \{(Y_{ij}^{(\ell)} | \mathbb{N}_{ij}^{(\ell)})\}_{i=1}^n$
 Obtain the pseudo MLE $\tilde{a}_j^{(\ell)}$ and $\tilde{b}_j^{(\ell)}$ based on $\mathbf{D}^{(\ell)}$
 Generate data $\mathcal{D}^{(\ell)} = \{(\mathcal{Y}_{ij}^{(\ell)} | \mathbb{N}_{ij}^{(\ell)})\}_{i=1}^n$ with $\mathcal{Y}_{ij}^{(\ell)}$ simulated from the null distribution $f_{\text{BB}}(y; \mathbb{N}_{ij}, \tilde{a}_j^{(\ell)}, \tilde{b}_j^{(\ell)})$ for every i .
 Calculate the test statistic $\Lambda^{(\ell)}$ based on $\mathcal{D}^{(\ell)}$
 Estimated p -value = $\frac{\#\{\ell: \Lambda^{(\ell)} \geq \Lambda\} + 1}{B + 1}$

We determine that taxon k needs the zero-inflation model if the corresponding Benjamini-Hochberg (BH) (Benjamini and Hochberg, 1995) corrected LRT p -value is less than 0.05. We refer to the GDM-based distribution with zero inflation determined this way as AIGDM.

3.3. AIGDM Marginal Model and Expectation-Solution Algorithm. Suppose we have N subjects and each subject has the composition of $K + 1$ microbial taxa measured over T time points. For notational simplicity, T does not vary across subjects, but our method does not require this. Let Y_{itj} denote the compositional count of taxon j measured at time t in the subject i . We assume $\mathbf{Y}_{it} = (Y_{it1}, \dots, Y_{it(K+1)})$ marginally follows the AIGDM distribution with the zero-inflated taxa indexed by set $\mathcal{K}_0$. Let Δ_{itj} be the latent absence/presence indicator for taxon j at time t in the subject i . The Δ_{itj} takes the value of 1 with probability π_{itj} for any $j \in \mathcal{K}_0$ and Δ_{itj} value is fixed at 0 for any $j \in \mathcal{K}_0^c$. Given $\Delta_{itj} = 0$, Z_{itj} is the latent beta variables with the mean $\mu_{itj} = a_{itj} / (a_{itj} + b_{itj})$ and dispersion $\phi_{itj} = 1 / (a_{itj} + b_{itj} + 1)$, where a_{itj} and b_{itj} are two shape parameters in the beta distribution. In the regression model, we link μ_{itj} , ϕ_{itj} , and π_{itj} with the design matrices $\mathbf{X}_{it}^{(\mu)}$, $\mathbf{X}_{it}^{(\phi)}$, $\mathbf{X}_{it}^{(\pi)}$

$$(3.2) \quad \mu_{itj} = \frac{e^{\alpha_j^T \mathbf{X}_{it}^{(\mu)}}}{1 + e^{\alpha_j^T \mathbf{X}_{it}^{(\mu)}}}, \quad \phi_{itj} = \frac{e^{\beta_j^T \mathbf{X}_{it}^{(\phi)}}}{1 + e^{\beta_j^T \mathbf{X}_{it}^{(\phi)}}}, \quad \pi_{itj} = \frac{e^{\gamma_j^T \mathbf{X}_{it}^{(\pi)}}}{1 + e^{\gamma_j^T \mathbf{X}_{it}^{(\pi)}}.$$

Let $\boldsymbol{\theta}$ denote the collection of all parameters $\boldsymbol{\alpha} = \{\alpha_j\}_{j=1}^K$, $\boldsymbol{\beta} = \{\beta_j\}_{j=1}^K$, and $\boldsymbol{\gamma} = \{\gamma_j\}_{j \in \mathcal{K}_0}$. For independent data, the EM algorithm can be used to efficiently estimate $\boldsymbol{\theta}$. However, the EM does not account for sample correlations in longitudinal studies. To address this problem, we adapt the Expectation-Solution (ES) algorithm (Rosen et al., 2000; Choo-Wosoba et al., 2016) in which the M-step of the EM algorithm is replaced by a step solving a set of GEE.

The complete data log-pseudo-likelihood of AIGDM is given by

$$\begin{aligned}
l(\mathbf{Y}, \mathbf{Z}, \mathbf{\Delta}; \boldsymbol{\theta}) &= \log \left[\prod_{i=1}^N \prod_{t=1}^T \left\{ f(\mathbf{Y}_{it} \mid Z_{it1}, \dots, Z_{itK}) \prod_{j=1}^K f(Z_{itj}) \right\} \right] \\
&= \sum_{i=1}^N \sum_{t=1}^T \log \{f(\mathbf{Y}_{it} \mid Z_{it1}, \dots, Z_{itK})\} \\
&\quad + \sum_{j=1}^K \sum_{i=1}^N \sum_{t=1}^T \left\{ \Delta_{itj} \log \pi_{itj} + (1 - \Delta_{itj}) \log(1 - \pi_{itj}) + \right. \\
&\quad \left. (1 - \Delta_{itj}) [-\log(\mathcal{B}(a_{itj}, b_{itj})) + (a_{itj} - 1) \log(Z_{itj}) + (b_{itj} - 1) \log(1 - Z_{itj})] \right\},
\end{aligned}$$

where $\mathcal{B}(\cdot, \cdot)$ is the beta function, $a_{itj} = \mu_{itj}(1/\phi_{itj} - 1)$ and $b_{itj} = (1 - \mu_{itj})(1/\phi_{itj} - 1)$. In the h -th iteration of the EM algorithm, we compute the expectation of the log-likelihood function $Q(\boldsymbol{\theta}) = E(l(\mathbf{Y}, \mathbf{Z}, \mathbf{\Delta}; \boldsymbol{\theta}) \mid \mathbf{Y}, \boldsymbol{\theta}^{(h)})$, with respect to the posterior distributions of $\mathbf{\Delta}$ and $\mathbf{Z}$ given $\mathbf{Y}$, using the current estimates of the parameters $\boldsymbol{\theta}^{(h)}$. The $Q(\boldsymbol{\theta})$ is then maximized to find the next parameter estimates $\boldsymbol{\theta}^{(h+1)}$. In the ES algorithm, instead of maximizing $Q(\boldsymbol{\theta})$, we solve a set of estimating equations. To prepare notation, we define $Z_{itj}^* = \log \frac{Z_{itj}}{1 - Z_{itj}}$ and $Z_{itj}^\dagger = \log(1 - Z_{itj})$. We have $\mu_{itj}^* = E(Z_{itj}^* \mid \Delta_{itj} = 0) = \psi(a_{itj}) - \psi(b_{itj})$, $\mu_{itj}^\dagger = E(Z_{itj}^\dagger \mid \Delta_{itj} = 0) = \psi(b_{itj}) - \psi(a_{itj} + b_{itj})$, $\text{Var}(Z_{itj}^* \mid \Delta_{itj} = 0) = \psi_1(a_{itj}) + \psi_1(b_{itj})$, where $\psi(\cdot)$ is the digamma function and $\psi_1(\cdot)$ is the trigamma function.

In the h -th E-step, we compute the following posterior means

$$\Delta_{itj}^{(h)} = E(\Delta_{itj} \mid \mathbf{Y}_{it}, \boldsymbol{\theta}^{(h)}) = \begin{cases} 0 & \text{if } j \in \mathcal{K}_0^c \text{ or } Y_{itj} > 0 \\ \frac{\pi_{itj}^{(h)}}{\pi_{itj}^{(h)} + (1 - \pi_{itj}^{(h)}) \frac{B(a_{itj}^{(h)}, b_{itj}^{(h)})}{B(a_{itj}^{(h)}, b_{itj}^{(h)})}} & \text{if } j \in \mathcal{K}_0 \text{ and } Y_{itj} = 0, \end{cases}$$

$$Z_{itj}^{*(h)} = E(Z_{itj}^* \mid \mathbf{Y}_{it}, \Delta_{itj} = 0, \boldsymbol{\theta}^{(h)}) = \psi(a_{itj}^{(h)}) - \psi(b_{itj}^{(h)}),$$

$$Z_{itj}^{\dagger(h)} = E(Z_{itj}^\dagger \mid \mathbf{Y}_{it}, \Delta_{itj} = 0, \boldsymbol{\theta}^{(h)}) = \psi(b_{itj}^{(h)}) - \psi(a_{itj}^{(h)} + b_{itj}^{(h)}),$$

where $a_{itj}^{(h)} = a_{itj}^{(h)} + Y_{itj}$ and $b_{itj}^{(h)} = b_{itj}^{(h)} + Y_{it(j+1)} + \dots + Y_{it(K+1)}$.

Let $\mathbf{\Delta}_{ij}^{(h)} = \{\Delta_{itj}^{(h)}\}_{t=1}^T$, $\mathbf{Z}_{ij}^{*(h)} = \{Z_{itj}^{*(h)}\}_{t=1}^T$, $\mathbf{Z}_{ij}^{\dagger(h)} = \{Z_{itj}^{\dagger(h)}\}_{t=1}^T$, $\boldsymbol{\pi}_{ij} = \{\pi_{itj}\}_{t=1}^T$, $\boldsymbol{\mu}_{ij}^* = \{\mu_{itj}^*\}_{t=1}^T$, $\boldsymbol{\mu}_{ij}^\dagger = \{\mu_{itj}^\dagger\}_{t=1}^T$, and $\boldsymbol{\phi}_{ij} = \{\phi_{itj}\}_{t=1}^T$ be the column vectors. In the h -th S-step, for each taxon j , we update the parameters $\boldsymbol{\alpha}_j$, $\boldsymbol{\beta}_j$, and $\boldsymbol{\gamma}_j$ by solving the estimating equations defined below.

Equations for $\boldsymbol{\alpha}_j$:

$$(3.3) \quad \boldsymbol{\Psi}_{\boldsymbol{\alpha}_j} = \sum_{i=1}^N \left\{ \frac{\partial \boldsymbol{\mu}_{ij}^*}{\partial \boldsymbol{\alpha}_j} \right\}^T \mathbf{V}_{Z_{ij}^*}^{-1} \mathbf{W}_{ij}^{(h)} \left\{ \mathbf{Z}_{ij}^{*(h)} - \boldsymbol{\mu}_{ij}^* \right\} = \mathbf{0},$$

where $\mathbf{W}_{ij}^{(h)} = \text{diag}\{1 - \Delta_{i1j}^{(h)}, \dots, 1 - \Delta_{iTj}^{(h)}\}$, $\mathbf{V}_{Z_{ij}^*} = \mathbf{D}_{Z_{ij}^*}^{1/2} \mathbf{R}(\rho_j) \mathbf{D}_{Z_{ij}^*}^{1/2}$, $\mathbf{D}_{Z_{ij}^*} = \text{diag}\{\text{Var}(Z_{i1j}^* \mid \Delta_{i1j} = 0), \dots, \text{Var}(Z_{iTj}^* \mid \Delta_{iTj} = 0)\}$, and $\mathbf{R}(\rho_j)$ is the working correlation matrix.

Equations for $\boldsymbol{\beta}_j$:

$$(3.4) \quad \boldsymbol{\Psi}_{\boldsymbol{\beta}_j} = \sum_{i=1}^N \left\{ \frac{\partial \boldsymbol{\phi}_{ij}^{-1}}{\partial \boldsymbol{\beta}_j} \right\}^T \left\{ \boldsymbol{\mathcal{M}}_{ij}(\mathbf{Z}_{ij}^{*(h)} - \boldsymbol{\mu}_{ij}^*) + (\mathbf{Z}_{ij}^{\dagger(h)} - \boldsymbol{\mu}_{ij}^\dagger) \right\} = \mathbf{0},$$

where $\mathcal{M}_{ij} = \text{diag}\{\mu_{i1j}, \dots, \mu_{iTj}\}$.

Equations for γ_j :

$$(3.5) \quad \Psi_{\gamma_j} = \sum_{i=1}^N \left\{ \frac{\partial \pi_{ij}}{\partial \gamma_j} \right\}^T \mathbf{V}_{\Delta_{ij}}^{-1} \left\{ \Delta_{ij}^{(h)} - \pi_{ij} \right\} = \mathbf{0}$$

where $\mathbf{V}_{\Delta_{ij}} = \mathbf{D}_{\Delta_{ij}}^{1/2} \mathbf{R}(\sigma_j) \mathbf{D}_{\Delta_{ij}}^{1/2}$, $\mathbf{D}_{\Delta_{ij}} = \text{diag}\{\pi_{i1j}(1 - \pi_{i1j}), \dots, \pi_{iTj}(1 - \pi_{iTj})\}$, and $\mathbf{R}(\sigma_j)$ is the working correlation matrix.

In this article, we employ compound symmetry structured $\mathbf{R}(\rho_j)$ and $\mathbf{R}(\sigma_j)$. The correlation coefficients ρ_j and σ_j are estimated through a separate set of GEE (section A.1 of the Supplementary Material). Alternative structures can also be considered. As in regular GEE, the misspecification of the working correlation does not affect the consistency of the parameter estimate. The detailed updating scheme for the ES algorithm is provided in section A.2 of the Supplementary Material. It follows from [Rosen et al. \(2000\)](#) that the estimators at convergence are consistent and asymptotically normal. Because the parameters for individual taxa are updated independently, we can estimate parameters for all taxa in parallel by distributing the optimization jobs across multiple computing cores.

3.4. Association Tests. By testing the different sets of parameters in the AIGDM model, we can reveal a variety of differential patterns in microbial compositions. We propose to perform three generalized score tests ([Rotnitzky and Jewell, 1990](#); [Boos, 1992](#)) to detect differential mean, dispersion, and proportion of absence/presence, respectively. As these tests focus on different aspects of the microbial composition distribution, they are complementary to each other. To achieve robustness and improve overall power, we construct an omnibus test by combining the p -values of the three tests.

When testing differential mean, we have the exposure variable (e.g., delivery mode) in $\mathbf{X}_{it}^{(\mu)}$ and the null hypothesis $H_0^{(\mu)} : \alpha_{11} = \dots = \alpha_{K1} = 0$, where α_{j1} is the parameter of interest in α_j for the exposure. To construct the score statistic for testing the null hypothesis, we calculate the restricted estimators of θ by fitting the reduced model under $H_0^{(\mu)}$ via the aforementioned ES algorithm. All the statistics below are evaluated at the restricted estimators. Let Ψ_α be the collection of all Ψ_{α_j} and Ψ_β be the collection of all Ψ_{β_j} . The score statistic for testing $H_0^{(\mu)}$ is

$$\mathbf{U} = \Psi_\alpha^{(1)},$$

where $\Psi_\alpha^{(1)}$ includes the components of Ψ_α corresponding to $\alpha_{11}, \dots, \alpha_{K1}$. Under the null, $\mathbf{U}$ is asymptotically multivariate normal with mean $\mathbf{0}$. The covariance matrix of $\mathbf{U}$ can be estimated using the adjusted sandwich formula

$$\mathbf{V} = \mathbf{B}^T \mathbf{A} \mathbf{B},$$

$$\mathbf{B} = \begin{bmatrix} \mathbf{I} \\ -\mathcal{I}_{12} \mathcal{I}_{22} \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \sum_{i=1}^N \left(\sum_{t=1}^T \Psi_{it} \right) \left(\sum_{t=1}^T \Psi_{it} \right)^T,$$

where $\mathbf{I}$ is a K -dimensional identity matrix, $\Psi = \{\Psi_\alpha, \Psi_\beta\}$, Ψ_{it} is the it -th summand in Ψ with the components for $(\alpha_{11}, \dots, \alpha_{K1})$ organized to appear first, $\mathcal{I}$ is the first derivative of Ψ with respect to (α, β) , $\begin{bmatrix} \mathcal{I}_{11} & \mathcal{I}_{12} \\ \mathcal{I}_{21} & \mathcal{I}_{22} \end{bmatrix}$ is the partition of $\mathcal{I}$ for $(\alpha_{11}, \dots, \alpha_{K1})$ vs the other parameters. Finally, we can construct the quadratic statistic for testing $H_0^{(\mu)}$

$$T = \mathbf{U}^T \mathbf{V}^{-1} \mathbf{U},$$

which asymptotically follows a chi-square distribution with K degrees of freedom.

The asymptotic approximation of the test statistic may not be accurate in the presence of rare taxa and when the sample size is low. In this case, we can use Smith’s permutation techniques (Winkler et al., 2014) to obtain more accurate p -values. As we are detecting the subject-level differences between exposure groups, the permutation is performed at the subject level. It is computationally efficient to obtain the permutation p -values for score test because the null model needs to be fit only once in the permutation procedure. To further increase computation speed, we employ an adaptive procedure that uses small numbers of permutations for large p -values and large numbers of permutations only for small p -values. In addition, we adopt a tail approximation strategy (Knijnenburg et al., 2009) to accurately estimate extremely small p -value without using a very large number of permutations. The algorithm of the adaptive permutation procedure is provided in section A.3 of the Supplementary Material.

We can construct the score tests to detect differential dispersion and differential proportion of absence/presence in a similar manner (section A.4 of the Supplementary Material). We employ the Cauchy p -value combination method (Liu and Xie, 2019) to combine the three p -values and obtain an omnibus test as

$$Q_{\text{AIGDM}} = \sum_{i=1}^3 w_i \tan[(0.5 - p_i)\pi],$$

where p_1 , p_2 , and p_3 are the p -values for testing differential mean, dispersion, and proportion of absence/presence, and w_i ’s are nonnegative weights with $\sum_i w_i = 1$ which are set to 1/3 in our implementation. The p -value of the AIGDM omnibus test can be approximated by $0.5 - \arctan(Q_{\text{AIGDM}})/\pi$. If we do not model zero inflation in any taxa in the composition, then the omnibus test only combines the differential mean test p -value and the differential dispersion test p -value.

4. Simulation Studies. We conducted extensive simulation studies to investigate the performance of the proposed and existing methods. In particular, we evaluated the proposed omnibus test when zero inflation is not modeled in any taxa (referred to as the GDM test), modeled in all taxa (referred to as the ZIGDM test), or adaptively determined (referred to as the AIGDM test). We also considered the DM-based omnibus test for detecting both differential mean and dispersion (La Rosa et al., 2012), as well as two distribution-free tests, QCATC1 and QCATC2, based on one-part and two-part models, respectively (Tang et al., 2017). We evaluated both the asymptotic and permutation versions of each test.

4.1. Simulation Strategy. We simulated $N = 50$ subjects, each with $T = 5$ (small sample size) or 20 (large sample size) repeated measures of compositional counts of 6 taxa (i.e., $K + 1 = 6$). Subjects were randomly assigned to the case ($X_{it} = 1$) and control ($X_{it} = 0$) groups with equal probability. We generated the compositional counts $\mathbf{Y}_{it}$ from GDM-based distributions. We considered settings where there is no zero inflation in any taxa or have zero inflation at either taxon 1 (the most abundant taxon) or taxa 2 to 5, with the baseline zero-inflation level set to 40%. The mean (μ_{itj}), dispersion (ϕ_{itj}), and proportion of absence/presence (π_{itj}) parameters were calculated based on equation (3.2), with the specific values of the regression coefficients summarized in Table 1. The values for α_0 and β_0 in Table 1 were obtained from fitting the GDM to a species composition under genus *Clostridium* in the ECAM data. For zero-inflated taxa, the values for γ_0 were set to -0.41 such that the baseline zero-inflation level is 40%. The α , β , and γ values in Table 1 determine the extent of the difference in mean, dispersion, and proportion of absence/presence between the case and control groups. In the type I error simulation, they were set to 0 for all taxa. In the power

TABLE 1
Summary of parameter setup in GDM-based simulations

Taxon	α_0	α	β_0	β	γ_0			γ
					no ZI	Taxon 1 ZI	Taxa 2-5 ZI	
1	-0.84		0.17		$-\infty$	-0.41	$-\infty$	
2	-1.04		1.18		$-\infty$	$-\infty$	-0.41	
3	-0.01	$\{0, 1, 0.8\}^\dagger$	0.74	$\{0, 0.8, 0.6\}^\dagger$	$-\infty$	$-\infty$	-0.41	$\{0, 1.6, 1\}^\dagger$
4	0.06		0.83		$-\infty$	$-\infty$	-0.41	
5	0.61		1.54		$-\infty$	$-\infty$	-0.41	

† Values in the parentheses are for different simulation settings.

simulation, we perturbed the mean, the dispersion, or the proportion of absence/presence of the active taxa. In the differential mean model, we randomly selected two active taxa and set their α values to 1 when $T = 5$ and to 0.8 when $T = 20$, while keeping their β and γ values at 0. In the differential dispersion model, we randomly selected two active taxa and set their β values to 0.8 when $T = 5$ and 0.6 when $T = 20$, while keeping their α and γ values at 0. In the differential absence/presence proportion model, we randomly selected one active taxon among the taxa with zero inflation and set its γ value to 1.6 when $T = 5$ and 1 when $T = 20$, while keeping its α and β values at 0.

Following (3.1), we generated compositional count $\mathbf{Y}_{it}$ using the (zero-inflated) beta variables. The sequencing depth $\mathbb{N}_{it}$ was randomly sampled from a Poisson distribution with a mean of 1000. For each subject i and taxon j , we simulated the correlated (zero-inflated) beta variables $Z_{i1j}, \dots, Z_{iTj}$ using the “NORmal to Anything (NORTA)” approach (Cario and Nelson, 1997):

1. If taxon j was not chosen to have the zero-inflation model, we set $\Delta_{itj} = 0$ for all t . If taxon j is chosen, we sampled $(Q_1, \dots, Q_T)$ from a T -dimensional normal distribution with zero means, unit variances, and correlation ρ . Then, we set $\Delta_{itj} = 0$ if $\Phi(Q_t) \leq 1 - \pi_{itj}$, where $\Phi(\cdot)$ is the cumulative distribution function for the standard normal distribution; otherwise set $\Delta_{itj} = 1$.
2. We sampled $(G_1, \dots, G_T)$ from a T -dimensional normal distribution with zero means, unit variances, and correlation ρ . We set $Z_{itj} = (1 - \Delta_{itj})F^{-1}(\Phi(G_t))$, where $F^{-1}(\cdot)$ is the quantile function of the beta distribution with mean μ_{itj} and dispersion ϕ_{itj} .

The $(Z_{i1j}, \dots, Z_{iTj})$ are correlated (zero-inflated) beta variables with their correlation dictated by ρ . We varied ρ from 0 to 0.8 in our simulation studies. For each simulation setting, we generated 5,000 datasets to evaluate type I error and 2,000 datasets to evaluate power at the nominal significance level of 0.05.

In addition, we performed another set of simulation studies where the compositional counts were generated from the logistic normal multinomial (LNM)-based distributions. The strategy for parameter setup and zero inflation configuration are similar to those in GDM-based simulations. The details are provided in section B of the Supplementary Material.

4.2. Simulation Results. Figure 2 displays the percentage of simulation replicates in which the zero inflation (ZI) state was determined correctly for each taxon under different zero inflation settings in GDM-based data-generating models. We can correctly determine no ZI most of the time, while the chance of correctly determining ZI depends on where the ZI is located. The chance is higher when ZI is at the abundant taxon (ZI at Taxon 1 setting) than when ZI is at the relatively low abundant taxa (ZI at Taxa 2-5 setting). The accuracy of classifying ZI increases with a higher sample size.

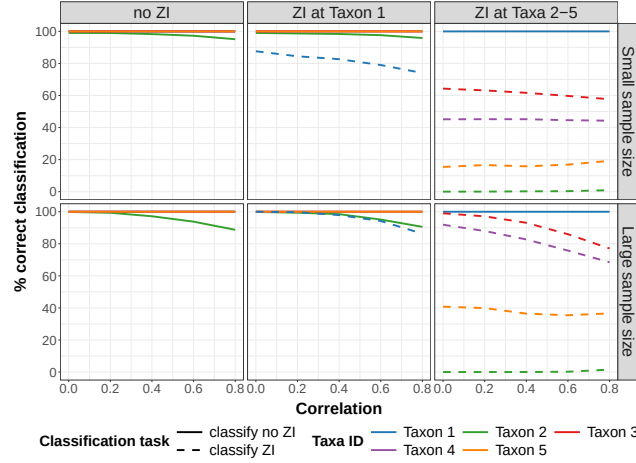


Fig 2: Percentage of simulation replicates in which the LRT accurately classifies the zero-inflation (ZI) state for each taxon.

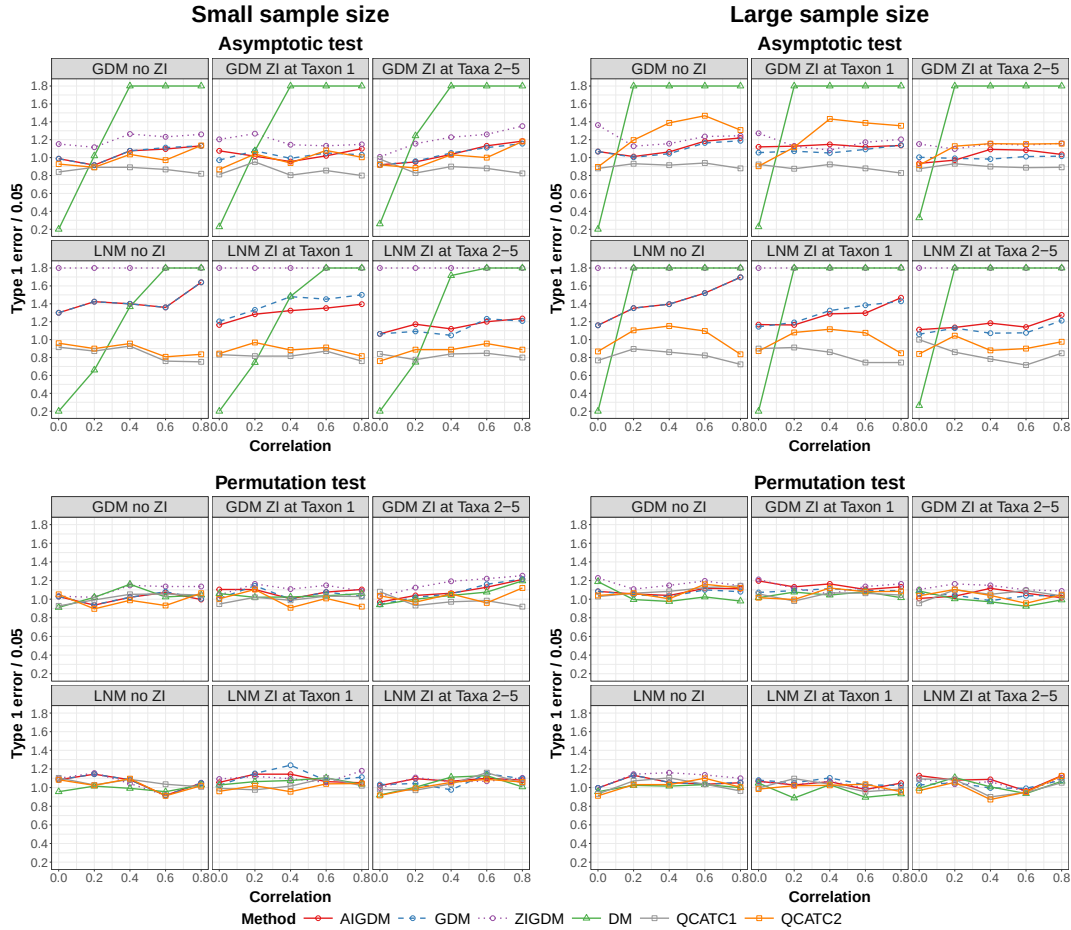


Fig 3: Empirical type I error rate divided by the significance level (0.05) of the asymptotic and permutation tests for detecting differential composition under small sample size setting (Left) and large sample size setting (Right). Some values for DM and ZIGDM asymptotic tests exceed the y-axis range but are positioned at the top for illustration purposes.

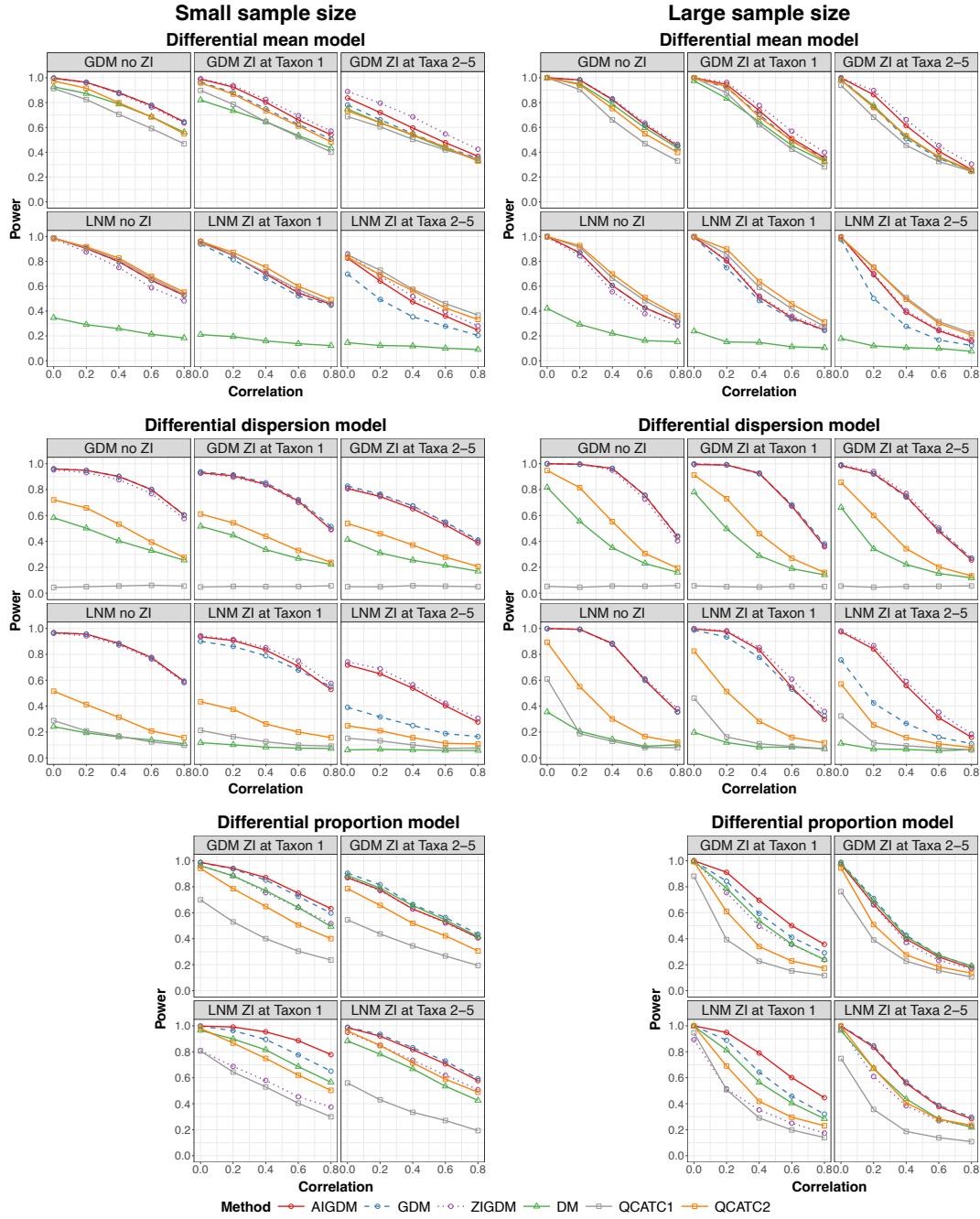


Fig 4: Power of the permutation tests for detecting differential composition under small sample size setting (Left) and large sample size setting (Right). Three differential patterns were considered (Upper: differential mean; Middle: differential dispersion; Lower: differential absence/presence proportion). The significance level is 0.05.

Figure 3 displays the empirical type I errors of the asymptotic and permutation tests. The asymptotic AIGDM and GDM control type I errors in GDM-based simulations and have inflation in LNM-based simulations. The asymptotic ZIGDM test has higher inflation than GDM and AIGDM tests. The asymptotic DM test has deflated type I errors when the intra-subject correlation is low and has inflated type I errors when the intra-subject correlation

is high. The asymptotic distribution-free tests (QCATC1 and QCATC2) and all permutation tests properly control type I errors.

For a fair comparison, we focused on the permutation tests in the power evaluation (Figure 4). All tests have decreased power when the intra-subject correlation increases. The AIGDM test has the highest power or comparable power to the most powerful method. This is true across all settings of differential patterns, zero inflation configurations, and data-generating models, demonstrating the robustness of the AIGDM test in enhancing statistical power. The GDM test has substantial power loss in the presence of ZI taxa, and the ZIGDM test has substantial power loss in detecting differential absence/presence proportion, especially when the data are generated from the LNM-based distributions. The QCATC1 and QCATC2 tests perform well in detecting differential mean but are less powerful in detecting differential dispersion or absence/presence proportion. The DM test has the lowest power in detecting differential mean or dispersion when the data are generated from the LNM-based distributions.

5. ECAM Data Analysis. We analyzed ECAM data introduced in Section 2 to identify differential compositions between C-section and vaginal birth infants. We aggregated OTUs into genera and excluded rare genera with a prevalence of less than 10%, resulting in 61 genera. These genera are further classified into 31 families, 18 orders, 12 classes, 6 phyla, and 1 kingdom (*Bacteria*). We applied the permutation tests evaluated in the simulations to analyze the 22 compositions defined at the internal nodes of the taxonomy tree (Figure 1c) and identify differential compositions while controlling the false discovery rate (FDR) at the 0.1 level using the BH procedure. The BH FDR-controlling procedure was also applied to the LRT test p -values for determining zero inflation across taxa in all compositions.

There were 11 compositions identified by at least one method (Table 2). The GDM-based omnibus tests (AIGDM/GDM/ZIGDM) identified 7 differential compositions, all of which are covered by the AIGDM omnibus test. The DM, QCATC1, and QCATC2 identified 4, 2, and 4 differential compositions. The compositions identified by the AIGDM test are highlighted in the taxonomic tree in Figure S1, and their p -values of the three tests for detecting differential mean, dispersion, and proportion of absence/presence are presented in Table S2. Most identified compositions have differential dispersion between C-section and vaginally delivered infants (Table S2). The genus composition under family *Ruminococcaceae* was uniquely detected by the AIGDM test to have the differential proportion of absence/presence between the two groups of infants (Table S2). The ZIGDM test indiscriminately models zero inflation in all taxa and yields less significant p -value compared to the AIGDM test (AIGDM p -value = 0.0242, ZIGDM p -value = 0.1079).

Over half of the compositions (55%) in this analysis were detected to exhibit zero inflation patterns in at least one taxon. Figure 5 illustrates an example of class composition under phylum *Proteobacteria*. The trajectories of the class relative abundance in the composition diverge between C-section and vaginally delivered infants over time (Figure 5a,b). Using the LRT, we detected one zero-inflated taxon (class *Betaproteobacteria*) in this composition and modeled zero inflation for the taxon in AIGDM. We further compared observed data and the synthetic data sampling from DM, GDM, and AIGDM distributions. In particular, we fit each distribution to the composition and randomly generated 5,000 synthetic datasets from each distribution with the estimated parameters. Our analysis demonstrates that the additional parameter to model zero inflation for class *Betaproteobacteria* in AIGDM is well-spent to provide a significantly better fit to the data (Figure 5c,d), producing a more powerful test (AIGDM p -value = 0.0051, GDM p -value = 0.0098, DM p -value = 0.0087). Another example of genus composition under family *Ruminococcaceae* is shown in Figure S2.

TABLE 2

The identified differential compositions between delivery modes in the ECAM data analysis. The table lists the name of the taxon at the parent node of the composition, the number of taxa in the composition (# Taxa), the number of taxa with detected zero inflation (# ZI), and the p -values of all the methods.

Composition	# Taxa (# ZI)	AIGDM	GDM	ZIGDM	DM	QCATC1	QCATC2
<i>f_Veillonellaceae</i>	4(0)	0.0315	0.0315	0.0503	0.1218	0.1584	0.1089
<i>f_Ruminococcaceae</i>	6(3)	0.0242	0.0692	0.1079	0.0528	0.2475	0.0419
<i>f_Enterobacteriaceae</i>	4(1)	0.0891	0.1091	0.0914	0.8728	0.0009	0.0299
<i>f_Clostridiaceae</i>	3(1)	0.0257	0.0271	0.1370	0.0023	0.2970	0.0396
<i>o_Pseudomonadales</i>	2(0)	0.0240	0.0240	0.0104	0.5257	0.5545	0.1078
<i>o_Lactobacillales</i>	4(0)	0.0636	0.0636	0.0685	0.0031	0.2178	0.0459
<i>o_Clostridiales</i>	6(2)	0.1862	0.1685	0.2072	0.0668	0.0519	0.0100
<i>o_Bacteroidales</i>	4(0)	0.1781	0.1781	0.3340	0.2653	0.3663	0.0068
<i>c_Actinobacteria</i>	2(1)	0.0144	0.0108	0.0144	0.0786	0.2574	0.0024
<i>p_Proteobacteria</i>	4(1)	0.0051	0.0098	0.0035	0.0087	0.2079	0.1198
<i>k_Bacteria</i>	6(0)	2.2e-5	2.2e-5	2.0e-5	2.6e-5	0.0011	0.0036

Note: Bold p -values are significant at FDR=0.1 level by applying BH to all p -values of each method. The label f_- , o_- , c_- , p_- , or k_- indicates the taxonomic level of the parent node of the composition (f_- : family, o_- : order, c_- : class, p_- : phylum, k_- : kingdom).

The AIGDM-identified compositions hold significant biological relevance, aligning well with recent research findings on the infant gut microbiome. Genus *Clostridium* under family *Clostridiaceae* and genus *Veillonella* under family *Veillonellaceae* was found to be more abundant in C-section born infants (Tamburini et al., 2016; Rutayisire et al., 2016), potentially due to increased exposure to maternal skin and the hospital environment where these genera are more prevalent (Coelho et al., 2021). On the other hand, order *Bifidobacteriales* under class *Actinobacteria* is likely transferred through the birth canal and tends to be more abundant and have higher colonization stability in vaginally delivered infants (Coelho et al., 2021). The genera *Faecalibacterium* and *Ruminococcus* under family *Ruminococcaceae* have also been reported to be more abundant in vaginally delivered infants (Busi et al., 2021).

6. Discussion. In this article, we introduce the AIGDM marginal regression model for longitudinal compositional counts in microbiome data exhibiting overdispersion and zero inflation. While existing zero-inflation models for microbiome data often consider observed zero counts as either true zeros arising from taxa absence (e.g., two-part models) or sampling zeros arising from the insufficient sampling of rare taxa (e.g., DM and GDM models), the AIGDM distribution allows for both types of zeros. We propose a LRT test to determine if modeling true zeros is necessary.

To efficiently estimate parameters in the AIGDM model with longitudinal data, we developed an ES algorithm that incorporates the GEE into the maximization step of the EM algorithm to account for intra-subject correlations. The ES algorithm is computationally efficient because of the simple calculation of posterior expectations and the ability to update parameters in a taxon-by-taxon manner. We propose three score tests to detect differential mean, dispersion, and proportion of absence/presence, and an robust omnibus test by combining the p -values of the three tests.

In a real data application, we demonstrate that the AIGDM fits the microbial composition well and is adaptable to various scenarios where there is no zero inflation or where there is zero inflation in a subset of taxa in the composition. The AIGDM tests powerfully identified differential gut microbial compositions between C-section and vaginally delivered infants and shed light on what aspects of the composition distribution was disrupted.

We focus on association analysis for compositions with a small number of taxa and utilize the quadratic statistic in association testing. However, when the number of taxa is large,

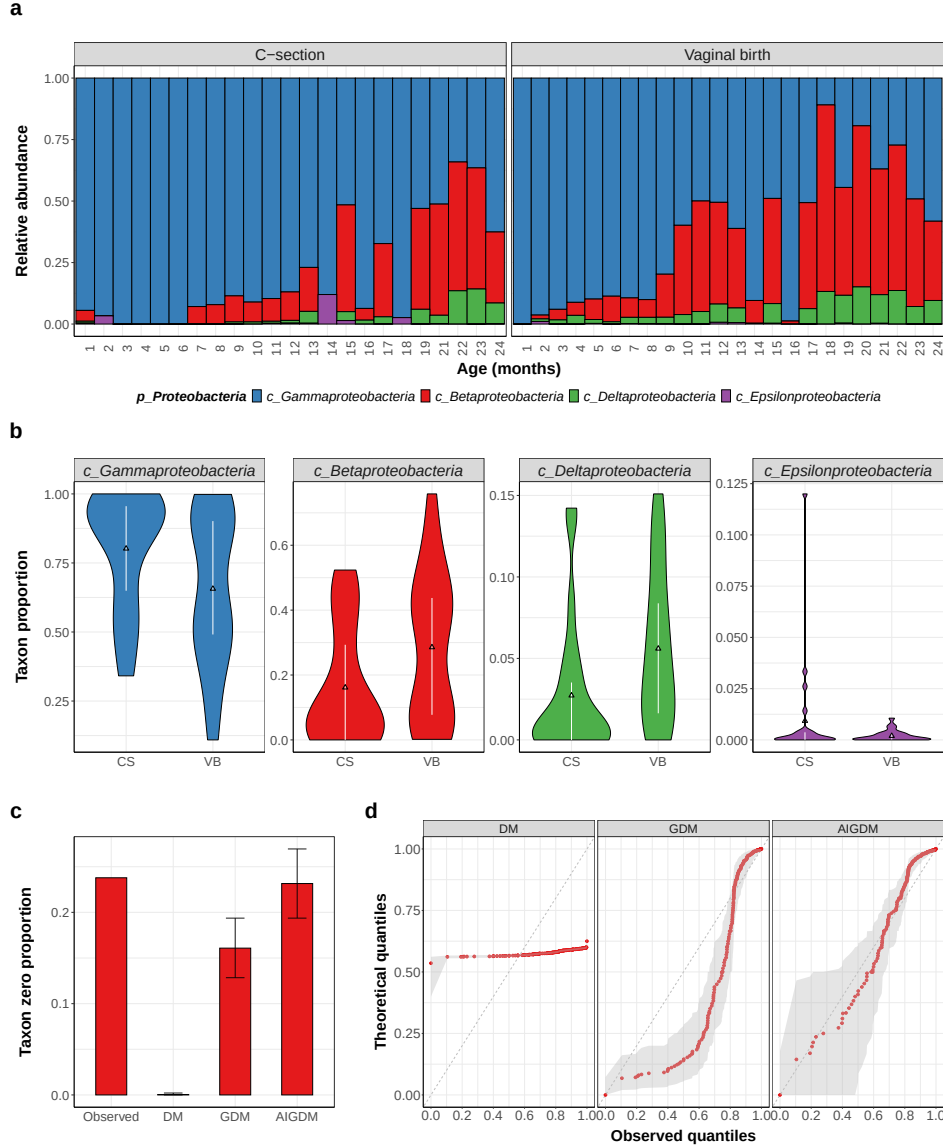


Fig 5: Information on class composition under phylum *Proteobacteria* in ECAM data. (a) Composition trajectory over two years of life for the C-section and vaginally delivered infants. Each bar represents the average composition across all samples measured in each month. (b) Violin plot comparing the distribution of the proportion of each taxon between the two groups. The white vertical line represents the interquartile range, and the black triangle indicates the mean. The plots in (c) and (d) focus on class *Betaproteobacteria* detected as a zero-inflated taxon. (c) Bar plot comparing the proportion of zero counts in the observed data and the synthetic data. The error bar is 95% confidence interval. (d) Quantile-quantile plot comparing synthetic data to observed data. Points on the plots represent medians, and the boundaries of the envelopes are 2.5 and 97.5 percentiles of the synthetic data.

the high degrees of freedom of the test can compromise its statistical power. In this case, other types of multivariate tests, such as the maximum test statistic and variance-component test statistic (Tang and Chen, 2021), can improve statistical efficiency. Similarly, different

p -value or test statistic combination strategies can be utilized to further improve the power of the omnibus test under certain differential patterns.

Our method has broad applications. In addition to taxa composition, it can also be applied to the composition of microbial genes or pathways to study the functionality of the microbiome and its relevance to the exposure of interest. Zero-inflated compositional counts are also prevalent in other omics data types generated from high-throughput sequencing, such as single-cell RNA-seq.

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SUPPLEMENTARY MATERIAL

In the Supplementary Material, Section A outlines GEE for correlation coefficients, provides more details of the ES algorithm and the adaptive permutation test algorithm, and describes the score test for differential dispersion and differential absence/presence proportion; Section B provides the details of the LNM-based simulation strategy; Section C contains additional results for the ECAM data analysis. R codes to implement the methods have been incorporated into the software miLineage, available at <https://tangzhengl.github.io/tanglab/software.html>. The scripts generating reported results are deposited at <https://github.com/KiRinHong/AIGDM>.

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